

Software Test Plan (STP) - Collaborative Study Material & Quiz Platform

Project: Collaborative Study Material & Quiz Platform

Version: 1.0

Authors: Team 10

Date: 13-09-2026

Status: Draft / Review

1. Introduction

Purpose: This document defines the test plan for the Collaborative Study Material & Quiz Platform v1.0. It outlines objectives, scope, strategy, resources, schedule, and responsibilities for testing.

Scope: Testing covers authentication, study materials, question bank, quiz creation/search, quiz attempt/evaluation, results/history, subject-wise performance, administration, security, performance, reliability, compatibility, and usability.

References: Collaborative Study Material & Quiz Platform SRS v1.5, Detailed Work Division for a 4-Member Software Engineering Team, UML specifications.

Definitions: SRS (Software Requirements Specification), STP (Software Test Plan), UI (User Interface), API (Application Programming Interface), MCQ (Multiple Choice Question), RTM (Requirements Traceability Matrix).

2. Test Items

- Authentication module
- Study Material Management module
- Question Bank Management module
- Quiz Creation and Search module
- Quiz Attempt and Evaluation module
- Results, History and Performance module
- Administrator/Admin and Security interface

3. Features to be Tested

Features mapped to SRS requirement IDs:

- CSMQ-F-001, F-002, F-004: Registration, login and logout
- CSMQ-F-010, F-012, F-013, F-041: Study-material upload, search/view, download and admin removal

- CSMQ-F-020, F-022, F-042: Question-bank add, search/filter and admin question/quiz removal
- CSMQ-F-030, F-032: Quiz creation and search/view
- CSMQ-F-033, F-034, F-035, F-036: Quiz attempt, evaluation, results and history
- CSMQ-F-040: Subject-wise average score
- CSMQ-NF-001: Response time within 3s
- CSMQ-NF-002: Data persistence after restart
- CSMQ-NF-003: At least 10 concurrent sessions
- CSMQ-NF-004: Supported browser compatibility
- CSMQ-NF-005: Clear and understandable UI feedback

4. Features Not to be Tested

- Enterprise/distributed-scale infrastructure beyond the single local/development deployment
- Specialized hardware interfaces (none are required by the SRS)
- Functionality not defined in the approved SRS

5. Test Approach / Strategy

Levels:

- Unit tests (module-level)
- Integration tests (frontend, REST APIs, backend, MongoDB)
- System tests (end-to-end Student and Administrator functionality)
- Acceptance tests (UAT)

Types:

- Functional testing (core features)
- Regression testing
- Performance testing (response time, concurrent users)
- Usability testing (UI clarity, navigation, feedback)

Entry Criteria: Stable build delivered, test data available, services and test environment ready.

Exit Criteria: 100% of planned test cases executed, High-priority requirements verified, no unresolved critical defects, all acceptance criteria satisfied.

5.1 Security Validation

- Validate password hashing and confirm no plain-text password storage
- Verify authentication before protected pages, functions and APIs
- Verify role-based access control for Administrator-only operations
- Validate and safely handle malformed or malicious input
- Verify HTTPS for sensitive communication in the deployed environment

6. Test Environment

Hardware: Desktop/laptop/tablet/smartphone devices; no specialized hardware required.

Software: React.js frontend, Node.js, Express.js REST API, MongoDB, modern web browsers.

Tools: Postman (API), browser developer tools, Jira (defect tracking), GitHub (test evidence/PRs).

Test Data: Student and Administrator accounts, valid/invalid credentials, PDF files, MCQs, quizzes, attempts and results.

7. Test Schedule

Milestones:

- Sprint 0: Test planning and environment setup
- Sprint 1: Authentication, PDF upload, Question Bank smoke testing
- Sprint 2: Material search/download, quiz creation/search, attempt/evaluation testing
- Sprint 3: History, performance, security, persistence, browser/usability and concurrency testing
- Sprint 4: Cross-testing, regression, final integration and UAT

8. Test Deliverables

- Test Plan (this document)
- Test Cases (manual & automated)
- Test Scripts / API test collections
- Test Data
- Test Execution Logs
- Defect Reports
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
QA Lead	Harshit Chandak	Prepare plan, coordinate execution
Test Engineer	Team 10	Design & execute test cases, log defects
Developer	Team 10	Support defect fixes and triage
Product Owner	Project Guide	Approve test results, sign-off readiness

10. Risks and Mitigation

Risk	Mitigation
------	------------

Delay in stable build delivery	Request early smoke builds from development team
Test environment or service downtime	Maintain backup test setup and restore procedures
Integration/API changes cause retesting delays	Coordinate contracts early and run regression smoke tests

11. Assumptions & Dependencies

- MongoDB test database and application services will be stable and available
- Required Student/Administrator test accounts and sample data will be available before execution
- The integrated MERN build and required browser versions will be available for testing

12. Suspension & Resumption Criteria

Suspend testing if:

- Test environment is unavailable for a material portion of the test window
- Build instability blocks more than 30% of planned test cases

Resume testing if:

- Blocking defects are resolved or formally accepted
- Environment is stabilized and smoke tests pass

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases.

Example:

- CSMQ-F-001 (Registration) -> TC-Auth-01
- CSMQ-F-030 (Quiz creation) -> TC-QUIZ-01
- CSMQ-NF-001 (Response time) -> TC-Perf-01

14. Test Metrics & Reporting

Metrics collected:

- % test cases executed
- % passed/failed
- Defect density
- Defect aging
- Requirement coverage

Reports:

- Daily execution status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
QA Lead	Harshit Chandak	
Dev Lead	Team 10	
Product Owner	Project Guide	